

CODE:

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main()
{
    int source, destination;
    char buffer[1024];
    ssize_t bytesRead;

    source = open("sample.txt", O_RDONLY);

    if (source == -1)
    {
        printf("Error opening source file.\n");
        return 1;
    }

    destination = open("copy.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (destination == -1)
    {
        printf("Error creating destination file.\n");
        close(source);
        return 1;
    }

    while ((bytesRead = read(source, buffer, sizeof(buffer))) > 0)
    {
        write(destination, buffer, bytesRead);
    }

    close(source);
    close(destination);

    printf("File copied successfully.\n");
}
```

OUTPUT:

a)

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-02$ ./filecopy
File copied successfully.
```

b)

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-02$ strace -o trace.txt cat sample.txt
Operating Systems Lab
System Calls Practical
This file will be copied.
```

c)

```

bhardwaj@Pranathi:~/2520890182_OSSP/2520890182_Practical/Practical-02$ cat trace.txt
execve("/usr/bin/cat", ["cat", "sample.txt"], 0x7ffc154b4a28 /* 27 vars */) = 0
brk(NULL)                               = 0x65054d0a3000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x790c98979000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=19183, ...}) = 0
mmap(NULL, 19183, PROT_READ, MAP_PRIVATE, 3, 0) = 0x790c98974000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\0\0\0\0\220\243\2\0\0\0\0\0...", 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0...", 784, 64) = 784
fstat(3, {st_mode=S_IFREG|0755, st_size=2125328, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0...", 784, 64) = 784
mmap(NULL, 2170256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x790c98600000
mmap(0x790c98628000, 1605632, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x790c98628000
mmap(0x790c987b0000, 323584, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1b0000) = 0x790c987b0000
mmap(0x790c987ff000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1fe000) = 0x790c987ff000
mmap(0x790c98805000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x790c98805000
close(3)                                = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x790c98971000
arch_prctl(ARCH_SET_FS, 0x790c98971740) = 0
set_tid_address(0x790c98971a10)         = 1231
set_robust_list(0x790c98971a20, 24)     = 0
rseq(0x790c98972060, 0x20, 0, 0x53053053) = 0
mprotect(0x790c987ff000, 16384, PROT_READ) = 0
mprotect(0x65052ccda000, 4096, PROT_READ) = 0
mprotect(0x790c989b9000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0

```